Spatial and Spectral Analysis of Hand versus Foot Motor Imagery EEG: ICA-Based Preprocessing and CSP+LDA Classification for Subject S023

BCI4NAT Seminar Report, Summer Semester 2026

Dataset: S023_hand_vs_foot (PhysioNet EEG Motor Movement/Imagery Dataset, Task 4, imagined hand vs. foot movement)

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Individual contributions:

Both authors contributed equally to data preprocessing, classifier implementation, and analysis. S. Padmanabhan Babu led the Methods and Code sections; A. Kanickairaj led the Results, Discussion, and literature review. Both authors reviewed and approved the final report.

1 Introduction

BCI systems classify mental states from EEG. Attention, workload, emotion, and motor imagery are all examples: internal states with a measurable EEG signature but no external behaviour attached to them. This project focuses on motor imagery, the mental state of rehearsing a movement without performing it, e.g. imagining opening a fist rather than doing it. No muscle activity is involved, so strictly nothing physical happens, but the same cortical machinery that plans real movement is still engaged. That is exactly why it works as a control signal for brain–computer interfaces (BCIs). It matters most for people who cannot move at all, from spinal cord injury, ALS, or stroke. If a BCI can tell different imagined movements apart, that gives someone a way to communicate or control a device with no physical output needed. This project is a small, scaled-down version of that same classification problem.

The EEG signature most associated with motor imagery is event-related desynchronisation, or ERD: a drop in mu (roughly 8–13 Hz) and beta (roughly 13–30 Hz) power over sensorimotor cortex during the imagery period (Pfurtscheller & Lopes da Silva, 1999). It is a frequency-domain signature, not a time-domain one. That distinction actually decides which feature is appropriate. An ERP approach looks for a fixed, time-locked waveform after a stimulus. That fits a short, stereotyped response, but motor imagery is not that: it is sustained and self-paced, with no sharp onset to lock a waveform to. What stays consistent instead is a shift in ongoing rhythmic power across the whole imagery period, and a band-power measure like ERD is built to pick that up. ERD is also not instantaneous. It builds up gradually after the cue, stays suppressed while the imagery continues, and is followed by a beta rebound (event-related synchronisation) once the task ends (Pfurtscheller & Lopes da Silva, 1999). Resting mu/beta reflects a synchronised, idling cortical state; engaging the motor system, even just by imagining a movement, breaks that synchrony. Where exactly this shows up on the scalp depends on which body part is involved, because motor cortex is organised somatotopically. Different regions of the precentral gyrus map onto different body parts: the hand area sits more laterally, the foot/leg area closer to the midline. So telling hand imagery apart from foot imagery is not about decoding movement directly. It comes down to which patch of sensorimotor cortex desynchronises more. Common Spatial Patterns is the feature-extraction method built around exactly this idea, and it works well for two-class problems like this one (Müller-Gerking et al., 1999; Ramoser et al., 2000).

This project uses EEG recordings from participant S023 of the PhysioNet EEG Motor Movement/Imagery dataset (Schalk et al., 2004), specifically the imagined hand-versus-foot condition. We preprocess the data in EEGLAB, extract CSP features, classify with linear discriminant analysis in BCILAB, and evaluate with cross-validation. The research question:

> Can EEG signals recorded during imagined hand and foot movement be reliably distinguished using a CSP+LDA classifier, and does the spatial pattern of the underlying EEG

activity match what is expected from the somatotopic organisation of motor cortex?

Our hypothesis, based on the ERD literature above, is that hand and foot imagery are both classifiable and spatially distinct. Hand and foot imagery should produce different mu/beta ERD topographies: foot closer to the midline (around Cz), hand more over the lateral central electrodes (C3/C4). We also expect a CSP+LDA classifier trained on this data to beat the chance level of roughly 51% by a clear margin. If the underlying signal is real and spatially separable, it should be learnable even from a small number of trials.

2 Methods

2.1 Experimental Paradigm

The data come from the PhysioNet EEG Motor Movement/Imagery dataset, a large public collection of recordings from 109 participants performing motor execution and motor imagery tasks under the BCI2000 system (Schalk et al., 2004). Subject S023 completed the imagined hand-versus-foot condition, task 4 in the original study protocol. A target appears either at the top or the bottom of a screen: top means imagine opening and closing both fists, bottom means imagine moving both feet. Nothing is actually moved at any point. Each trial starts with the target's appearance, and the next cue follows exactly 8.2 s later. We confirmed this fixed interval directly from the recording's own event timestamps rather than assuming it. Runs alternate between the two conditions in a pseudo-randomised order, so participants cannot predict which task is coming next. Rest periods between runs let the participant recover attention before the next block.

2.2 Data Acquisition

EEG was recorded with a 64-channel montage on the international 10-10 system, using BCI2000 at a sampling rate of 160 Hz (Schalk et al., 2004), per the original PhysioNet documentation. The data for this project came already converted to EEGLAB format: a `.set` file (metadata, channel locations, event markers) plus a matching `.fdt` file (the raw continuous data).

2.3 Dataset

The file used, `S023_hand_vs_foot.set`, contains continuous EEG with three event marker types: `T0` for resting periods, `T1` for hand-imagery onset, `T2` for foot-imagery onset. The recording runs 369 s (6.2 min), three concatenated runs of 123 s each, joined at two `boundary` events, again confirmed directly from the event structure rather than assumed. Each run gives 15 trials at a fixed 8.2 s cue-to-cue spacing, 45 trials total: 23 hand, 22 foot. Close enough to 50/50 that we skipped any class-balancing step before training.

2.4 EEG Preprocessing

All preprocessing was done in EEGLAB. Parameters below follow the level of detail Keil et al. (2013) recommend for EEG studies: reference scheme, filter cutoffs, epoch window, and baseline correction are all stated explicitly, nothing left implicit. Order of operations: re-reference to the average of all channels; a zero-phase FIR filter combining a 7 Hz high-pass (removes slow drift below mu) with a 30 Hz low-pass (removes high-frequency noise above beta), run as one band-pass operation via `pop_eegfiltnew`; a 50 Hz notch filter for line noise; resample to 100 Hz; then ICA (extended infomax) on the continuous, filtered data. 7–30 Hz keeps mu and beta and drops everything else. 100 Hz is plenty, since nothing of interest here lives above 30 Hz.

Average-referencing a 64-channel recording costs one degree of freedom, so EEGLAB reported a data rank of 63 going into ICA, not 64. Expected, not a problem. It just means ICA returned 63 components instead of 64.

We inspected the resulting components by scalp topography, time course, and power spectrum. One complication is worth stating plainly: the data were already band-pass filtered to 7–30 Hz before ICA ran, so no component's spectrum can show the classic broadband or low-frequency-heavy shape a raw eye-blink or muscle artifact usually has outside that band. That information was gone before ICA ever saw the data. What we could still check is spectral shape *within* the 7–30 Hz band: narrow rhythmic peak, or flat and broadband. Figure 1 shows this for the four components discussed below.

On that basis we removed two. IC6 and IC15 both had a small, sharply-bounded topography over the left temporal area, unlike the smoother, broader pattern a cortical source usually shows, plus sharp, spike-like time courses rather than continuous brain-rhythm fluctuation. Both point to muscle (EMG) artifact. Their within-band spectra told a mixed story: IC15 is close to flat and broadband, which is the shape expected from muscle, so that supports removal directly. IC6 shows a moderate 10–11 Hz peak, so the spectral case alone is weaker there and the decision leans more on topography and time course.

We kept two others as genuinely brain-like. IC1 has a smooth, broad posterior topography and, by a clear margin, the sharpest peak of the four (around 10 Hz, a textbook posterior alpha rhythm), with a waxing-and-waning time course rather than sharp transients. That is the strongest spectral evidence in this set. IC4 sits over roughly the sensorimotor area, an anatomically sensible spot for a task-related component, with a time course equally free of spike-like transients. Its within-band spectrum is fairly flat though, so unlike IC1, the case for keeping it rests mainly on topography and time course, not spectral shape.

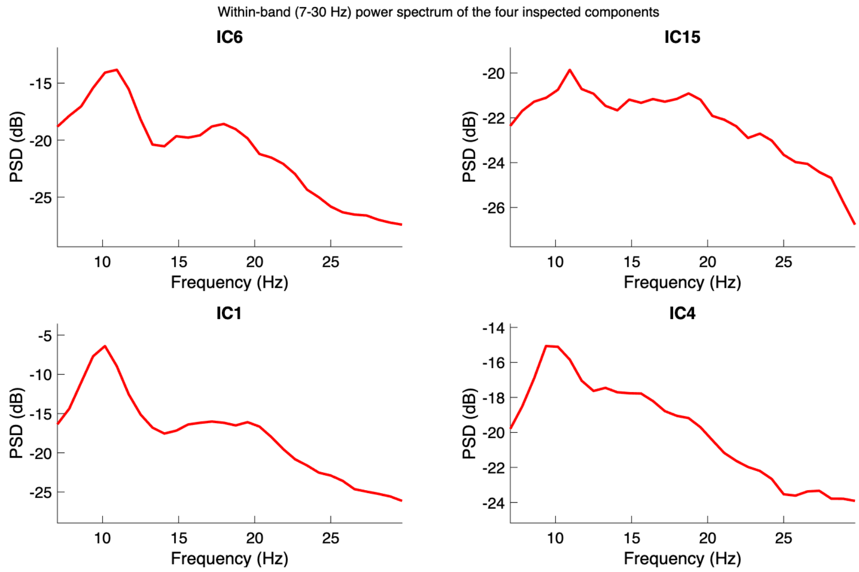

Figure 1: Within-band (7–30 Hz) power spectrum of the four inspected independent components, subject S023. IC1 (kept) shows a sharp, prominent peak near 10 Hz. IC15 (removed) is close to flat with no clear peak. IC6 (removed) and IC4 (kept) both show weaker, broader spectral structure, so for these two the topography and time-course evidence above carries more weight than spectral shape alone.

After component removal, the cleaned continuous dataset was saved separately, before epoching, as required for submission. Epochs were then extracted from -0.5 to 3.5 s relative to cue onset: half a second before the cue as baseline, three seconds of imagery after. Each epoch was baseline-corrected against the -500 to 0 ms window. Worth flagging honestly: an early version of our pipeline tried cropping each epoch down to just the post-cue analysis window, using EEGLAB's `pop_select` on already-epoched data. That silently corrupted the event labels, caught by checking trial counts before and after: 23 hand / 22 foot had dropped to essentially one usable trial. The fix was to keep the full epoch and take the analysis sub-window as a plain array slice later in the pipeline instead.

2.5 BCI Approach

At a high level, BCI classification turns a continuous, noisy brain signal into a single discrete decision: given one trial's EEG, output one label from a fixed set, here hand or foot. That is what a real BCI would act on, e.g. to move a cursor left or right. Raw voltage traces on their own are not a usable control signal. Getting from raw EEG to that decision takes two steps, feature extraction and classification, and BCILAB was used for both.

Feature extraction and classification were both done through BCILAB, using its `CSP` paradigm (Common Spatial Patterns) with linear discriminant analysis as the learner. In BCILAB, the whole pipeline is declared as one nested configuration block, an `Approach`, rather than as separate calls:

```
{'CSP', 'SignalProcessing', {'FIRFilter','off', 'Resampling','off',
```

```
'EpochExtraction', [0.5 3.5]}, 'Prediction', {'FeatureExtraction',
  {'PatternPairs', 3}, 'MachineLearning', {'Learner', 'lda'}}}
```

The outer `'CSP'` selects the paradigm. `SignalProcessing` controls what BCILAB does to the raw signal before feature extraction. `FIRFilter` and `Resampling` are explicitly turned off, since `ParadigmCSP` otherwise applies its own default filtering and resampling on top of data that is already filtered and resampled in EEGLAB; leaving them on would silently re-process the signal a second time. `EpochExtraction` is the only signal-processing step actually left to BCILAB, since referencing, filtering, and ICA were already handled in EEGLAB and re-running them would be redundant. `Prediction` nests the feature-extraction settings (three CSP pattern pairs) inside the classifier settings (LDA as the learner). This one block is what `bci_train` takes as its `Approach` argument, and it is where the feature-extraction/classifier choice actually gets made, not just described. The choice of feature follows straight from the EEG signature in the Introduction: motor imagery shows up as ERD, a change in mu/beta band power, and CSP is built for exactly that. It learns spatial filters, each one a weighted combination of the 64 channels. Push the data through a filter and you get a single signal whose variance should differ as much as possible between the two classes. Band power and signal variance are the same quantity here, both computed from the same band-passed 7–30 Hz signal. A filter that maximises variance for hand and minimises it for foot is picking out exactly the spatial pattern of band-power difference between the two conditions, which is the ERD contrast we are trying to classify. We used three CSP pattern pairs: six spatial filters, six log-variance features per trial, one number per filter. Fairly standard for a dataset this size; many more filters and you risk overfitting with only 45 trials.

LDA was the classifier, again through BCILAB (`ml_trainlda`), default regularisation. Given the six log-variance features for a trial, LDA finds a linear boundary through that six-dimensional space, chosen to best separate hand trials from foot trials during training. BCILAB's LDA does not hand back a bare label directly. For each trial it returns a discrete score per class, how strongly the trial's features favour hand versus foot, and the hard prediction is whichever score is higher: the argmax. The pooled confusion matrix in Section 3.2 was built the same way, argmax of each fold's per-trial scores rather than a pre-computed label. Either way the output ends up as one predicted label per trial, hand or foot, plus the fold-by-fold statistics in Section 3.2. We also tried plain, unregularised LDA out of curiosity; it made results slightly worse, not better, so we kept the default.

2.6 Classifier Training

The classifier was trained directly on the `T1` (hand) and `T2` (foot) markers, using BCILAB's `bci_train` with a 5-fold cross-validation scheme. With 45 trials split roughly evenly, each fold had around nine test trials. The classes were close to balanced (23 vs. 22), so the theoretical chance level is just the larger class's share: 23/45, 51.11%.

3 Results

3.1 EEG Patterns of Hand vs. Foot Motor Imagery

Before looking at classifier performance, it is worth checking the data actually contain the kind of signal motor imagery is supposed to produce, rather than just trusting the classification number on its own. Figure 3 shows event-related desynchronisation (ERD): percentage power drop during imagery relative to a 0.5 s pre-cue baseline, at three electrodes over the sensorimotor strip, C3, Cz, and C4. We used 0.5 to 3.5 s after the cue as the imagery window, to capture the sustained part of the ERD response rather than its onset. ERD takes a moment to build up, and these trials run long enough for that. Figure 3 pools ERD over that whole 3 s window; Figure 2 breaks it down into three sub-windows to show how it develops over time.

The clearest difference is at Cz, mu band (8–13 Hz): hand shows strong desynchronisation there, around 83% ERD, while foot is weaker, around 75% (higher ERD% means a bigger power drop from baseline). That is the opposite of what our hypothesis predicted. We expected a stronger foot signal at Cz, given the medial position of the leg/foot representation in motor cortex (Pfurtscheller & Lopes da Silva, 1999). C3 and C4 show a smaller, noisier version of the same pattern, and neither condition consistently wins there. Figure 4 shows the same thing as scalp topography: in the mu band, the foot map has a localised teal/blue dip right over the central midline, while hand stays strongly desynchronised (yellow) across nearly the whole scalp, including that spot. So it is foot imagery that shows the localised weak point at the midline, not hand. The beta-band maps look more similar between conditions, which fits the smaller, noisier C3/C4 difference above.

Figure 2 splits the 0.5–3.5 s window into three roughly one-second sub-windows, early/mid/late, and repeats the same baseline-relative mu-band ERD% at Cz for each. Hand rises from 79.8% early to a peak of 88.0% mid-window, then falls back to 80.2% late. Foot is highest early, 82.1%, then drops and stays lower: 73.5% mid, 73.7% late. Neither condition is flat and sustained across the window. Hand builds and peaks mid-imagery; foot is strongest right after the cue and fades. The pooled full-window numbers in Figure 3, then, are an average over a response that changes shape over time.

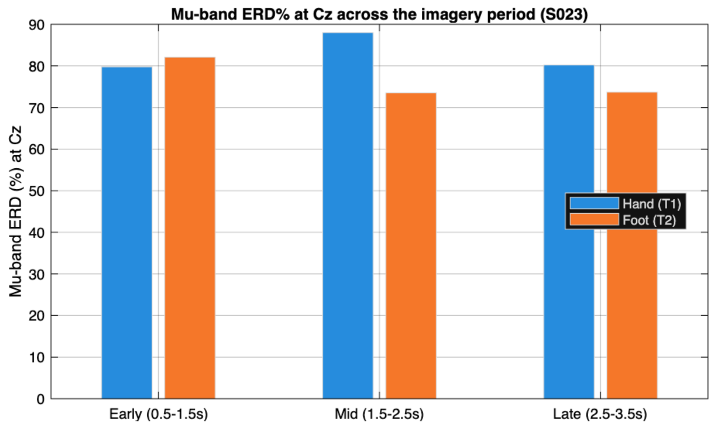

Figure 2: Mu-band (8–13 Hz) ERD% at Cz, split into early (0.5–1.5 s), mid (1.5–2.5 s), and late (2.5–3.5 s) sub-windows of the imagery period, hand versus foot, subject S023.

One limitation is worth stating clearly rather than glossing over. The absolute ERD percentages here, roughly 65–100% depending on channel and frequency, are on the high end even for this task. Pfurtscheller and Lopes da Silva (1999) note that ERD magnitude is sensitive to exactly this kind of methodological choice: the percentage is relative to a baseline period, so it depends on how reliably that baseline power gets estimated. We traced ours to the short baseline window, 0.5 s, only 51 samples at the 100 Hz rate used here, which makes the baseline power estimate noisier and biased upward. A related bug, mismatched FFT window sizes between the baseline and imagery calculations, made it worse before we caught it and partially fixed it. Values came down from a physiologically implausible 90–100% to a still-elevated but more believable range. The hand-versus-foot pattern is the finding we actually care about, and that should still hold, since the same bias hits both conditions equally. The absolute percentages themselves, though, should not be compared directly to numbers from studies with longer baseline windows.

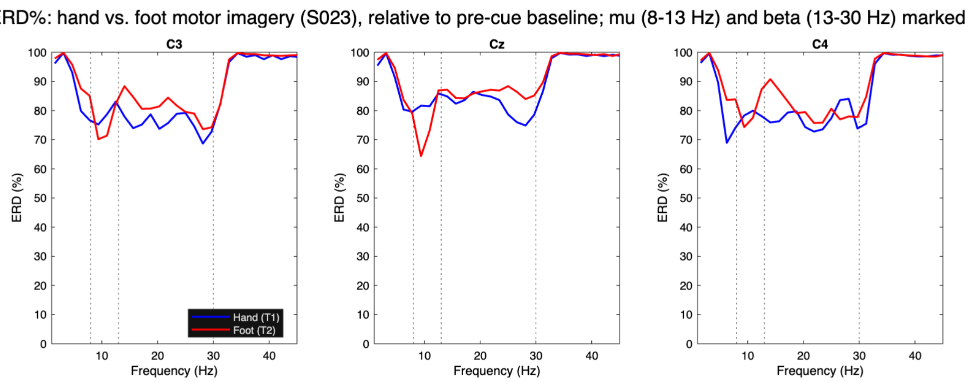

Figure 3: ERD% (percent power drop relative to a 0.5 s pre-cue baseline) at electrodes C3, Cz, and C4 for hand (blue) versus foot (red) motor imagery, subject S023. Vertical dotted lines mark the mu (8–13 Hz) and beta (13–30 Hz) band edges. Hand shows a stronger mu-band ERD% at Cz than foot does.

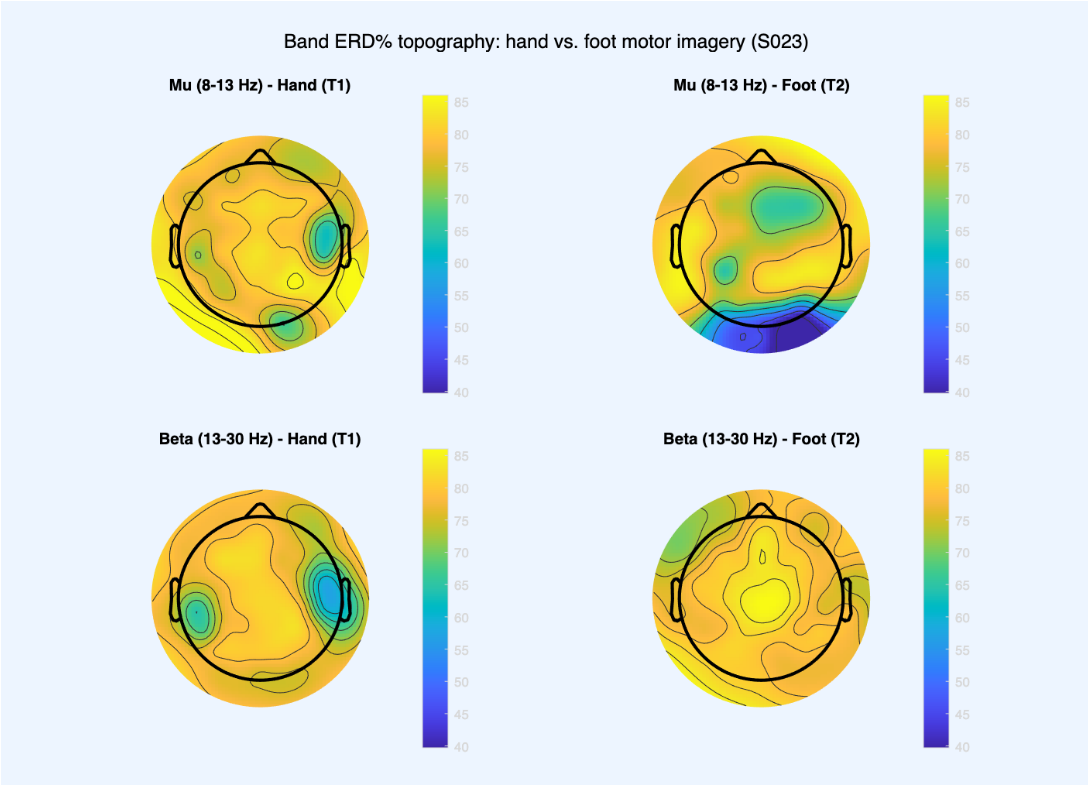

Figure 4: Scalp topography of mu- and beta-band ERD% for hand versus foot motor imagery, subject S023.

3.2 BCI Model Validation

The CSP+LDA classifier, trained and evaluated through BCILAB's `bci_train` with 5-fold cross-validation, reached 62.22% accuracy (misclassification rate 0.3778). Getting a trustworthy number here took more work than expected. BCILAB's fold assignment is supposed to be seeded internally by default, but it was not actually landing on the same split run to run, even with that default in place, so we stopped relying on it. Instead the 5-fold split is now generated in our own script with a seeded `randperm` and handed to `bci_train` as an explicit list of train/test index pairs. This takes BCILAB's internal fold-assignment logic out of the picture entirely. We confirmed it is exactly reproducible: six full reruns in a row, same fold indices and same confusion matrix every time.

The theoretical chance level, given the 23/22 split, is 51.11%. That is simply what a classifier gets for free by always predicting the larger class, computed straight from the class proportions with no need to touch the data. The empirical chance level is different: instead of being computed, it is estimated by repeatedly shuffling the class labels at random, retraining and re-evaluating on each shuffled version, and looking at the resulting spread of accuracies (a permutation test). Shuffling destroys any real link between the EEG features and the true class, so that spread shows what accuracy this exact pipeline would get by chance alone, given this data and procedure, not just the class split. It is a stricter baseline, one that can catch a pipeline that only beats the theoretical chance level because of some artifact in the data or procedure. Estimating it was optional here and was not run, so 51.11% is what the 62.22% result is measured against: 11.11 points above chance. Better than random, but not by a wide margin.

Table 1 shows the pooled confusion matrix across all 5 folds, built directly from each fold's per-trial predictions, not from averaged rates. Of 23 hand trials, 13 were correctly classified as hand and 10 as foot (56.5% correct). Of 22 foot trials, 15 were correct and 7 went to hand (68.2% correct). The classifier does somewhat better on foot than hand, though the two are closer together than in earlier debugging runs of this pipeline, where a stale random seed had produced a wider, less consistent split between the classes' accuracy. Table 2 summarises these rates alongside overall accuracy and chance level. Per-fold misclassification rates had a mean of 0.378 and a median of 0.333 (SD 0.169), the same source of spread as before, since each fold has only about nine test trials, so one wrong prediction shifts a fold's error rate by more than ten points.

Table 1: Pooled confusion matrix, 5-fold cross-validation, BCILAB CSP+LDA (all 45 trials, one prediction per trial from its held-out fold).

		Predicted	
		Hand	Foot
Actual	Hand	13	10
	Foot	7	15

Table 2: Classifier performance, 5-fold cross-validation, BCILAB CSP+LDA.

Metric	Value
Accuracy	62.22%
Theoretical chance level	51.11%
Hand trials correctly classified	56.5% (13/23)
Foot trials correctly classified	68.2% (15/22)
Hand trials misclassified as foot	43.5% (10/23)
Foot trials misclassified as hand	31.8% (7/22)
Misclassification rate, mean ± SD	0.378 ± 0.169

For comparison, we also implemented the same CSP+LDA pipeline directly in MATLAB, outside BCILAB's built-in functions, mainly as a sanity check on the BCILAB result. That version reached a noticeably higher accuracy, around 87% averaged across repeated random 5-fold splits. We tested class imbalance, fold randomness, and LDA regularisation settings directly, and none of them explained the gap. The actual cause: the two implementations estimate the class covariance matrices CSP is built on differently. BCILAB pools all trials of a class and computes one covariance matrix; our manual version computes a covariance per trial, normalises each one, then averages, closer to the original CSP formulation in Ramoser et al. (2000). Both are legitimate, just not numerically the same, especially with only 45 trials, and the two gave visibly different spatial filters when compared directly (essentially zero correlation between the weight sets). The assignment requires BCILAB functions specifically, so 62.22% is the figure we report as the main result. The manual pipeline's higher number is a cross-check here, not an alternative result to pick from.

Figure 5 shows the six spatial patterns (three pattern pairs) the BCILAB model learned. Most are focal rather than spread diffusely: patterns 1, 2, and 5 show tight, localised weighting over temporal and bilateral fronto-central regions. That points to the classifier picking up on localised activity, not some global artifact like a reference shift or slow drift. Patterns 3, 4, and 6 are more bilateral or spread across a wider region. Less textbook, but not unexpected. With only 45 trials, some of that is probably CSP fitting to whatever happened to separate the two classes best in this small sample, not purely the "true" motor-cortex signal. Not every pattern lines up neatly with the classic C3/Cz/C4 sites a larger, cleaner dataset would likely show more clearly. On balance we would call these patterns broadly consistent with a real neural signal, not a precise localisation of one.

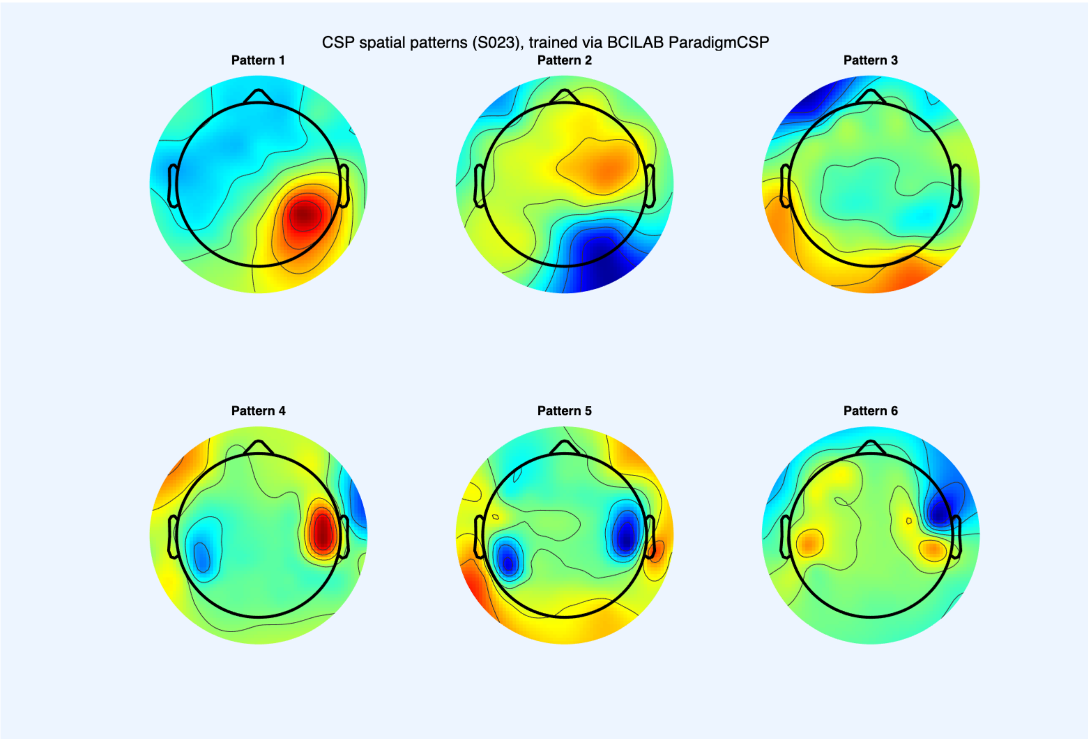

Figure 5: CSP spatial patterns (6 filters, 3 pattern pairs) learned by the trained model, subject S023.

4 Discussion

The research question asked two things: can EEG from imagined hand and foot movement be reliably told apart with a CSP+LDA classifier, and does the underlying spatial pattern match what somatotopic organisation of motor cortex would predict. First part: yes, with caveats. Second part: no. The spatial pattern we found runs opposite to what somatotopic organisation would predict for this subject.

On classification, the BCILAB CSP+LDA pipeline reached 62.22% accuracy against a chance level of 51.11%. Above chance, and that fits the first half of our hypothesis: a real, spatially separable signal should be learnable from this data even with only 45 trials. It is not a high accuracy by the standards of the motor-imagery literature, where 80–90% is common for two-class problems, but those studies typically have more trials per participant and sometimes more channels or a longer session to work with. Given the constraints here, 62.22% is a modest result, above chance, but only by around 11 points, a narrower margin than we expected going in.

On the spatial side, the ERD results in Section 3.1 do not support the second half of our hypothesis. We predicted foot would show a stronger effect near the midline (Cz) and hand a stronger effect laterally (C3/C4). Cz showed the opposite, hand's ERD% clearly beating foot's, and neither electrode showed the predicted lateral advantage for hand. A genuine miss, not a partial success. We would not read this as evidence against somatotopic organisation in general, since one subject and 45 trials is a thin basis for testing something as broad as cortical topography, but our specific prediction about where hand versus foot signals would be strongest was wrong, for this subject and this dataset.

A few limitations are worth stating explicitly, beyond what is already noted in Methods and Results. This is a single-subject analysis, so it says nothing about generalisation to someone else with different scalp anatomy, a different resting mu rhythm, or less consistency at the task. The ERD magnitudes in Section 3.1 are inflated by the short pre-cue baseline window, as discussed there; the direction and relative size of the hand-versus-foot difference should still be trustworthy, but the absolute percentages should not be read too literally. The gap between the BCILAB result (62.22%) and our manual CSP+LDA cross-check (around 87%) traces back to how the two implementations estimate class covariance matrices, and it is a reminder that "the accuracy" of a BCI pipeline depends on implementation choices that matter on a small dataset. 62.22% is what this implementation achieves, not a ceiling on the data. And several runs of what we thought was the same seeded pipeline (Section 3.2) produced different confusion matrices at first, because BCILAB's internal fold-seeding did not actually pin down the split as documented; generating it ourselves fixed that.

If we were continuing this project, the obvious next step is more trials, or data from more of the 109 PhysioNet participants, so the classifier is not trained and evaluated on such a small sample. A cross-subject classifier would also be more realistic, since a real assistive-BCI system ideally would not need retraining per user. Running the empirical chance level would give a more rigorous baseline, and re-running ICA before the band-pass filter would fix the diagnostic-information problem from Section 2.4.

As for a real-world application: the obvious one is the motivation from the Introduction, an assistive interface for people who cannot move but can still imagine movement. A hand-versus-foot classifier is a minimal example. The same approach scales to more imagined movements, which is the direction most practical BCI systems go in anyway.

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